

ISFAHAN SHAHID BEHESHTI INTL, Iran, Islamic Republic of

WMO: 408000

Lat: 32.751N

Lon: 51.862E

Elev: 1546

StdP: 84.08

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.2	(c) -6.8	(d) -18.2	(e) 0.9	(f) 3.9	(g) -16.2	(h) 1.1	(i) 4.8	(j) 11.5	(k) 12.5	(l) 9.9	(m) 10.1	(n) 1.6	(o) 270	(p) 0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.8	(c) 39.1	(d) 16.8	(e) 38.1	(f) 16.3	(g) 37.0	(h) 16.0	(i) 18.5	(j) 36.0	(k) 17.6	(l) 35.3	(m) 16.9	(n) 34.7	(o) 3.5	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	10.4	25.6	10.1	9.3	22.7	9.0	8.6	22.2	59.3	36.0	55.9	35.2	53.2	34.7	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.3	8.5	7.1	-11.6	40.6	2.0	0.8	-13.1	41.1	-14.2	41.6	-15.3	42.0	-16.8	42.6
			-12.5	19.9	1.7	2.0	-13.8	21.4	-14.8	22.6	-15.8	23.7	-17.0	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	2.5	5.4	10.2	15.2	20.7	26.0	28.6	26.5	22.6	16.6	8.4	4.0
	DBStd	9.40	3.34	3.56	3.53	3.05	2.73	2.25	2.16	2.14	2.19	2.64	3.50	3.00
	HDD10.0	664	233	132	41	2	0	0	0	0	0	1	72	186
	HDD18.3	2023	491	362	252	99	10	0	0	0	1	67	298	443
	CDD10.0	2714	1	3	48	159	331	481	576	511	377	203	23	1
	CDD18.3	1031	0	0	1	7	83	231	318	252	128	12	0	0
	CDH23.3	17141	0	0	26	199	1406	3675	5194	4089	2111	437	3	0
	CDH26.7	9795	0	0	3	31	568	2182	3344	2502	1069	96	0	0
Wind	WSAvg	2.9	2.3	3.0	3.5	3.6	3.7	3.4	3.1	2.8	2.7	2.6	2.2	2.1
Precipitation	PrecAvg	123	23	14	26	15	8	1	1	1	1	5	11	17
	PrecMax	196	66	45	66	39	27	9	11	3	3	36	39	40
	PrecMin	57	0	1	0	0	0	0	0	0	0	0	0	0
	PrecStd	38	16	13	18	12	8	2	2	1	1	8	13	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	20.0	26.9	29.7	34.2	39.1	40.2	39.8	35.2	30.8	23.2	17.9
		MCWB	6.2	7.7	10.3	12.4	14.3	16.3	17.4	16.5	14.9	13.1	10.0	7.8
	2%	DB	14.1	17.2	23.1	27.2	32.9	38.0	39.8	38.1	34.8	29.0	21.1	15.1
		MCWB	5.1	6.4	8.8	11.8	13.8	15.9	17.2	16.4	14.8	12.8	9.1	6.3
	5%	DB	12.1	15.8	21.0	25.8	31.2	36.8	38.9	37.0	33.6	27.8	19.0	13.1
		MCWB	4.4	5.9	8.1	11.2	13.4	15.4	16.8	16.2	14.4	12.3	8.6	5.2
	10%	DB	10.1	13.9	19.1	24.0	30.0	35.2	37.9	35.9	32.2	26.0	17.0	11.1
		MCWB	3.4	5.0	7.4	10.3	13.0	14.7	16.2	16.0	13.9	11.6	7.6	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	9.2	11.4	13.9	16.8	19.1	20.3	19.8	17.2	14.7	11.4	8.7
		MCDB	13.0	16.5	19.3	24.0	30.4	36.0	36.9	35.0	32.8	27.7	19.7	14.2
	2%	WB	6.5	7.8	10.2	12.9	15.2	17.6	18.8	18.0	15.9	13.5	10.2	7.6
		MCDB	11.7	14.6	19.5	23.4	30.1	34.5	36.7	35.3	32.2	26.7	17.8	12.7
	5%	WB	5.2	6.7	9.3	12.1	14.4	16.4	17.9	17.2	15.0	12.7	9.4	6.6
		MCDB	10.4	13.6	18.4	22.7	28.8	33.8	36.0	35.2	31.4	25.4	16.9	11.2
	10%	WB	4.0	5.5	8.3	11.4	13.6	15.5	17.1	16.4	14.3	11.9	8.4	5.4
		MCDB	9.2	12.5	17.3	21.6	27.4	33.0	35.2	34.5	30.7	24.1	15.7	9.8
Mean Daily Temperature Range	5% DB	MDBR	14.3	15.6	16.1	15.6	17.1	18.1	17.8	19.1	19.3	17.9	15.4	14.2
		MCDBR	16.9	17.6	18.4	17.6	18.7	19.6	18.6	19.3	20.1	19.5	17.9	16.8
		MCWBR	10.2	9.7	9.2	7.7	7.4	7.7	7.2	7.9	8.6	9.3	9.7	10.0
	5% WB	MCDBR	14.5	15.1	15.5	15.3	17.3	18.3	17.0	18.4	19.1	17.9	15.2	13.7
		MCWBR	9.2	9.0	8.2	7.0	7.2	7.8	7.2	8.1	8.8	8.9	8.5	8.7
Clear-Sky Solar Irradiance	taub	0.290	0.325	0.384	0.463	0.467	0.455	0.480	0.432	0.382	0.381	0.329	0.290	
	taud	2.447	2.319	2.108	1.915	1.924	1.967	1.921	2.052	2.179	2.156	2.333	2.463	
	Ebn at Noon	935	926	889	831	829	836	812	848	880	846	872	913	
	Edh at Noon	90	113	151	192	192	184	191	165	139	132	100	84	
All-Sky Solar Radiation	RadAvg	3.32	4.30	5.48	6.18	7.25	8.29	7.90	7.48	6.44	4.97	3.60	3.06	
	RadStd	0.22	0.27	0.30	0.40	0.37	0.26	0.21	0.17	0.16	0.20	0.27	0.25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-1.11	N/A	N/A	-1.24	N/A	N/A	+115	-98	-61
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.99	N/A	N/A	+115	-98	-61

Nomenclature: See separate page